

Software Architecture- 01

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ILOs

1. Use unified modelling language(UML) to document architectural design
2. Explain how architecture can be understood from multiple perspectives and through different UML diagrams.
3. Explain different software architectures, including:
 - when and how they are used
 - their advantages and disadvantages
 - how to represent them visually
4. Analyze and evaluate system architectures.

ILOs

Explain how software architecture is tied to:

- implementation languages
- deployment environments
- the structure of the development team
- the needs of the business and development cycle
- opportunities for product lines
- the intent of the software



Architect will consider ...

what will the building be used for?

Who will be using the building?

What qualities matter most to the occupants

What will happen if ?

What would happen if there was no clear design for the building?

Each work team would construct their own section and they would each have different views on what is needed

What if one team decides that accessibility is the most important factor to consider, while another team decides that security is the most important factor?

The two ideas do not work together and overall, it may not suit the needs of the users. You get conflicts, you get duplication

Software architecture

The software architecture of a system is the set of structures needed to reason about the system, which comprise software elements, relations among them, and properties of both

Software Architecture

Software architecture is the fundamental design of an entire software system.

It defines what elements are included in the system, what function each element has, and how each element relates to one another.

It is the big picture or overall structure of the whole system—how everything works together.

Software Architecture [cont..]

It is the big picture or overall structure of the whole system—how everything works together.

- the purpose of the system,
- the audience or users of the system,
- the qualities that are of most importance to users,
- where the system will run

Software architecture is important, particularly for large systems. If there is a clear design of the overall system from the start, there is a solid basis for developers to follow.

Advantages

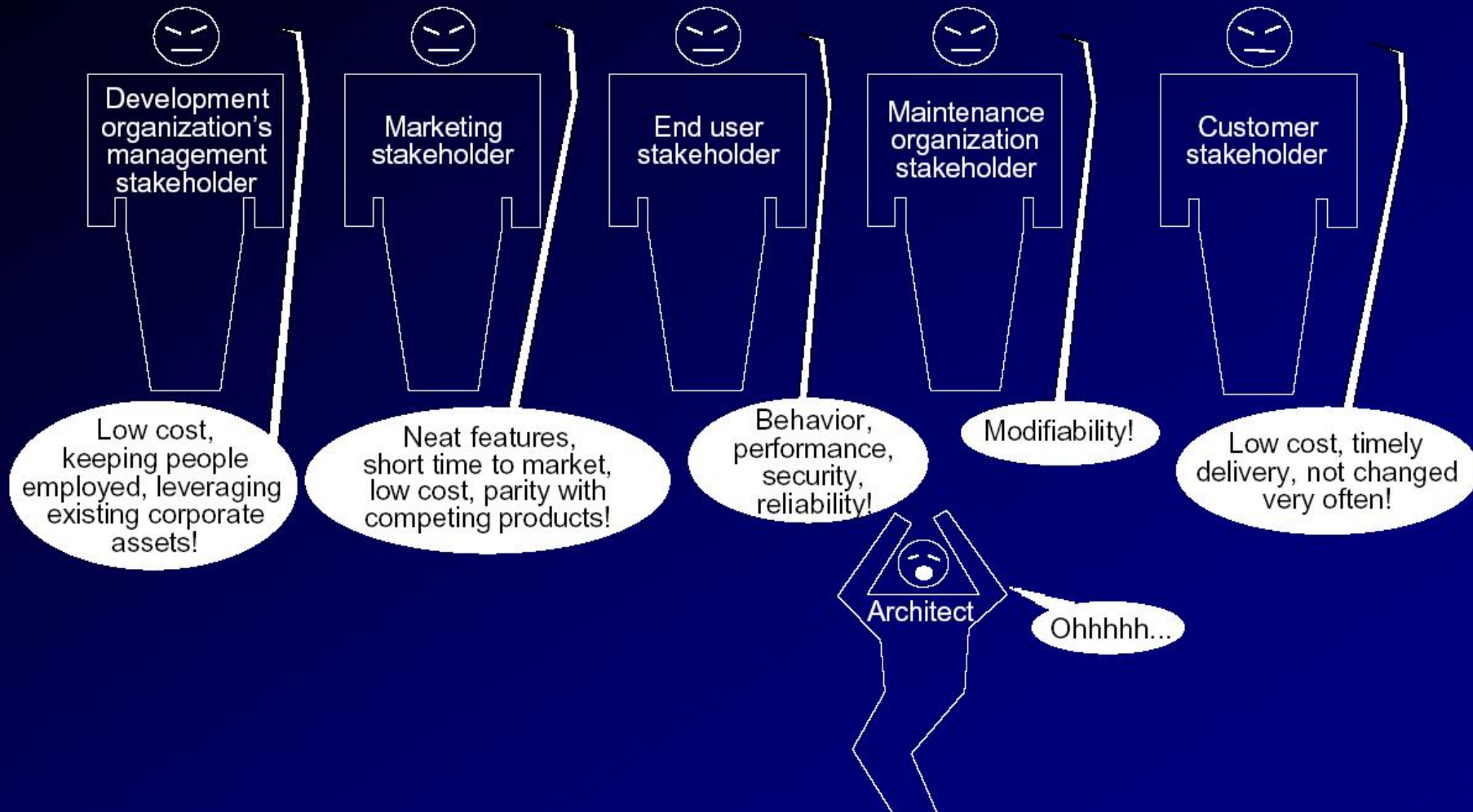
Some advantages of software architecture include:

- higher **productivity** for the software team, as a well-defined structure helps to coordinate work, implement individual features, or guide discussions on potential issues
 - improved **evolution** for the software, since design principles are applied to make changes easier to accomplish or defects easier to find
 - enhanced **quality** in the software by carefully considering the needs and perspectives of all the stakeholders
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- Software architecture helps make a system easier to maintain, reuse, and adapt.

How to develop?

In order to develop software architecture, architects must take into account the **stakeholders** of the system. Stakeholders are the people who have an interest in the software at hand. They either use the system or benefit from it in some way.

Stakeholders of a System



Stakeholder	Description
Software developers	Software architecture helps developers create and evolve software by providing strong direction and organization on what needs to be done.
Project Managers	Software architecture provides useful information to project managers to help them identify possible risks and to manage the project successfully. Software architecture helps project managers to understand task dependencies and impacts of change and to coordinate work assignments.
Clients	Clients make important decisions about the system, like its funding. Software architecture establishes a basis for communication with clients, so they understand what they are paying for and that their needs are met.
End users	Users may not care how the software is actually designed, but they do care that it “works well” for them.

Kruchten's 4+1 View Model

Multiple perspectives are necessary to capture the complete behaviour and development of a software system.

- Functionality of the software(**logical view**)
- How well the software executes. (**process view**)
- Implementation considerations (**Development view**)
- Physical considerations(**Physical view**)

Scenarios outline use cases or tasks required by the end users, which provides context to help to detail the four views.

Logical view

The logical view, which focuses on the functional requirements of a system, usually involves the objects of the system. From these objects, a UML class diagram can be created to illustrate the logical view.

Process view

The process view focuses on achieving non-functional requirements. These are the requirements that specify the desired qualities for the system, which include quality attributes such as performance and availability. The process view also presents processes that correspond to the objects in the logical view.

Development view

The development view describes the hierarchical software structure. It also considers elements such as programming language, libraries, and toolsets.

Essentially, the development view covers the hierarchical software structure and project management.

Physical View

The physical view handles how elements in the logical, process, and development views must be mapped to different nodes or hardware for running the system.

Scenarios

Scenarios align with the use cases or user tasks of a system and show how the four other views work together.

- For each scenario, there is a **script** that describes the sequence of interactions between objects and processes.
- This involves the key objects defined in the logical view, the processes described in the process view, the hierarchy identified in the development view, and the different nodes specified in the physical view. Scenarios relate these elements to provide a complete picture.